IE 643 COURSE PROJECT · PREP PRESENTATION

Probing the Latent Space of Video Diffusion U-Nets

Where does a video model keep what a thing looks like, and where does it keep how the thing moves?



Team Claudn't

Roll numbers — [fill in, one per team member]

This work is carried out as part of the IE 643 course project.

22 September 2026

Three questions, in order

Where the split should be, what we have measured so far, and what we do between now and the assessment.

I — Where would the split be, and who has looked?

- Content and motion, and the edit that tests them
- How a video diffusion U-Net is put together
- Activation patching, and cross-attention as a lens
- Where three recent papers already found motion

slides 3–8

II — What we have actually run and measured

- AnimateDiff, and a probing set we generate ourselves
- Five pilots: feasibility, seed control, storage, causality

slides 9–10

III — The five deliverables, and the plan behind them

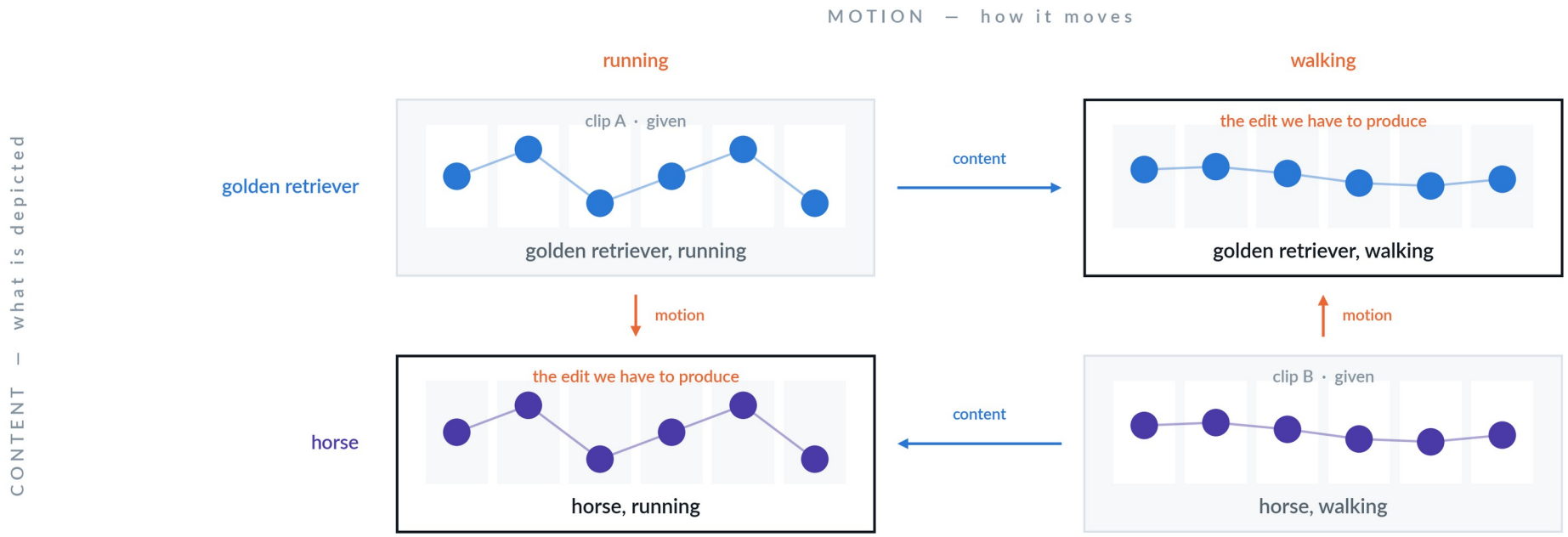
- Harness, selectivity map, patching, the demonstration, a LoRA
- Metrics and baselines, the frozen protocol, the roadmap, the risks

slides 11–22

Fifteen minutes. The background is short because the measurements are the interesting part.

Two things get settled at once, and we want to move just one

Content is what is depicted: identity, colour, texture, layout. Motion is what happens across frames: gait, deformation, the path traced through the frame. The test is an edit that changes one and leaves the other alone.



If we can say which layers and which denoising steps hold which, we can make that edit. That is the whole project.

The architecture already factors the problem — structurally

Spatial blocks see one frame at a time. Temporal blocks are the only path along which information crosses frames.

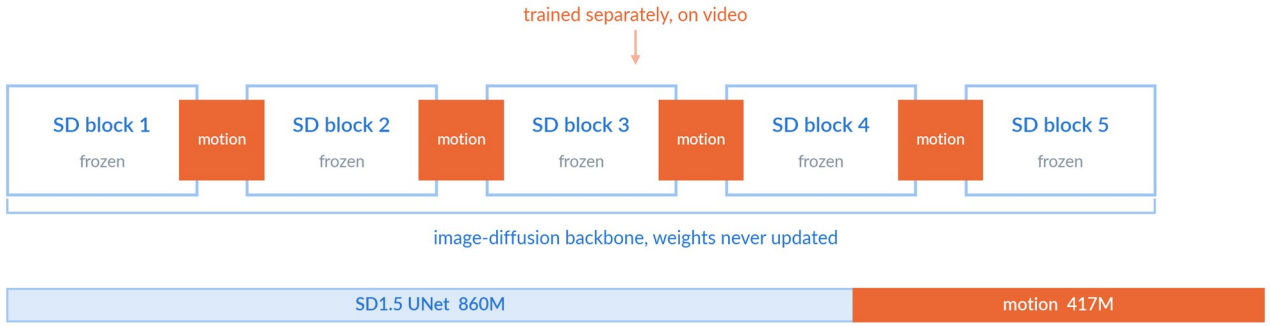


Our contribution is to show the split also holds causally — a different and much stronger claim than an architectural diagram.

Ho, NeurIPS 2022 · Blattmann, CVPR 2023 · Blattmann, 2023

AnimateDiff hands us the split as two separate parameter sets

It takes a pretrained image model, freezes it, and threads a separately trained motion module between its blocks. Content and motion live in different weights by construction — the handle we grab first.



Sizes, corrected

SD1.5 UNet	≈ 860M, frozen
Motion module	≈ 417M, trained
On disk	mm_sd_v15_v2, 1.6 GB
Peak VRAM	10.2 GB on a T4 (E1)

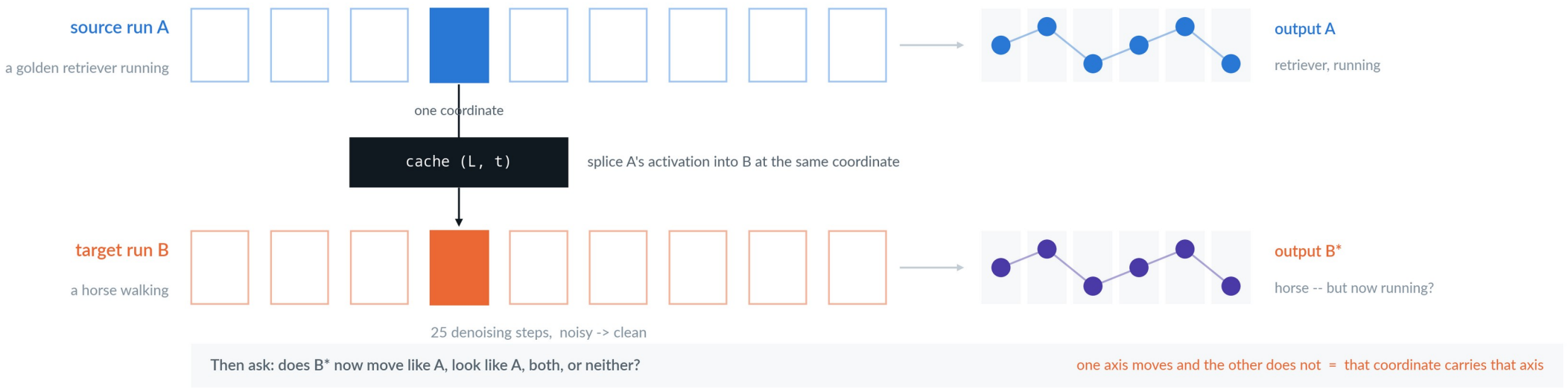
An earlier draft of ours said 1.7B. That was the file size, not a parameter count.

The catch we say before anyone asks. The backbone was frozen, so its layers never saw motion during training. Some of the separation is an artefact of that recipe — which is why a jointly-trained model stays on the list.

Guo et al., ICLR 2024 · Zhao et al., ECCV 2024 · Wu et al., ICCV 2023

Patching turns a hunch about a layer into a causal claim

Cache what one layer produced at one timestep on clip A, then splice it into clip B's run at the same coordinate.



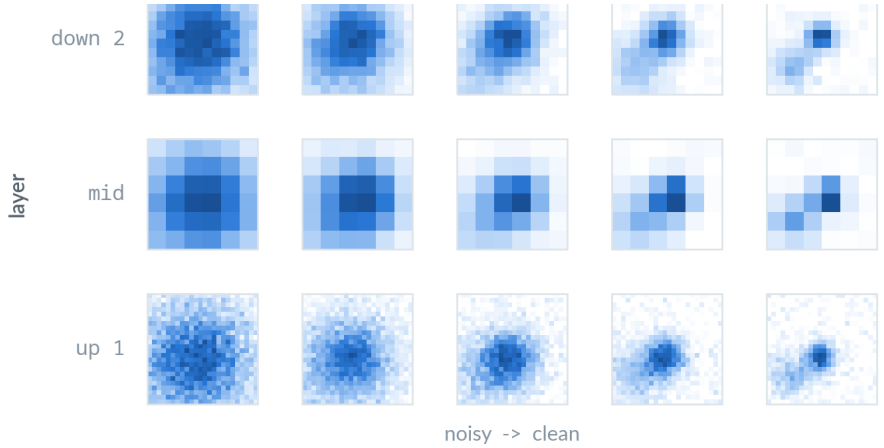
All three of these come from language-model interpretability and the causal logic carries over unchanged. We are reading Zhang and Nanda before writing any patching code: their warnings about metric choice, and about noising versus resampling corruption, describe mistakes we would otherwise make.

Meng et al., NeurIPS 2022

Wang et al., ICLR 2023

Zhang & Nanda, ICLR 2024

Cross-attention shows what a layer is looking at



DAAM aggregation with the frame axis kept instead of collapsed, so a token's map becomes a small movie.

Schematic. The real maps arrive with the Stage 2 harness.

A cross-attention map says which regions, and which frames, a text token is read into — at every layer and every denoising step.

Cheap

No intervention, no second generation. We can look before we commit to patching anything.

Per-token

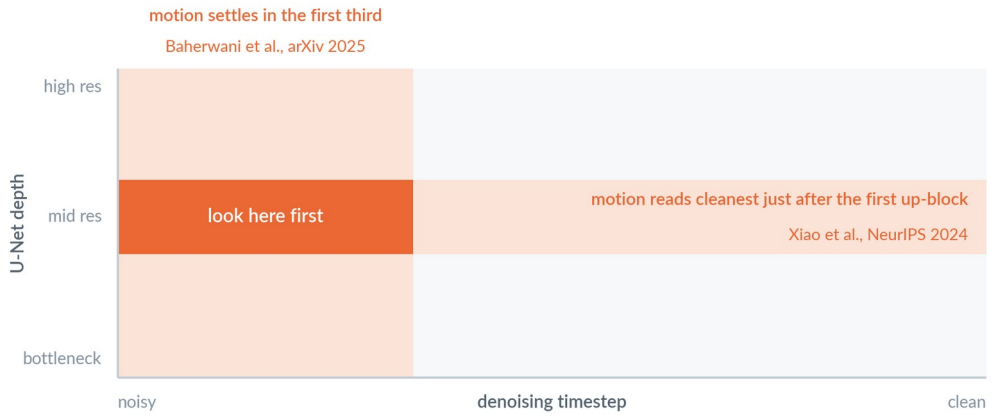
The map for “retriever” and the map for “running” are different objects, which is exactly the distinction we are chasing.

Kept as a movie

DAAM collapses the frame axis. We keep it, head-averaged and downsampled to 16 × 16.

Hertz et al., ICLR 2023 · Tang et al., ACL 2023

Three recent results tell us where to look first



And the signal sits in temporal attention, not in the features.

Ling et al., ICLR 2025 · Yatim et al., CVPR 2024

LAYER

Mid-resolution, not the extremes

Xiao et al. build a motion feature from U-Net activations and find the cleanest signal just after the first up-block. Deeper and shallower blocks both do worse.

TIMESTEP

Motion early, appearance late

Widely assumed, rarely tested. Baherwani et al. measured it and report two regimes with a reasonably stable boundary.

SIGNAL

The attention, not the features

MotionClone finds the dominant components of temporal attention drive motion; Yatim et al. get a motion descriptor by averaging space-time features over space.

And the older framing. MoCoGAN (2018) split content from motion by hand in the generator's latent code, and CMD (ICLR 2024) does the diffusion-era version. Both build the split in. We are asking whether a model that was never factored this way learns one anyway.

AnimateDiff is primary, and we generate every clip we probe

MODEL	SIZE	WHY IT IS ON THE LIST	WHY IT IS NOT PRIMARY
AnimateDiff PRIMARY	417M + 860M	Explicit spatial / temporal split; 10.2 GB measured on a T4; mature diffusers support	The motion module sits on a frozen image model, so the split is partly given
ModelScope T2V	1.7B	Spatial and temporal trained jointly, end to end. The harder test case	Thinner tooling for attention extraction than SD-derived pipelines
Stable Video Diffusion	≈ 1.5B	Best output quality of the three, well documented	Image-conditioned only, so matched text-prompt pairs are not possible

The probing set, frozen 22 September

15 subjects × 4 verbs × 2 scenes = 120 clips per seed; three seeds (42, 1337, 2024) gives 360.

Subjects split into two body-plan families — 8 quadrupeds, 7 humans — so a noun swap does not drag the layout with it.

104 within-family content pairs · 90 motion pairs · 56 cross-family, reported separately as a labelled hard subset.

What these models were pretrained on

SD1.5 backbone: LAION-5B subsets — web-scraped image–text pairs, with known demographic and aesthetic bias.

Motion module: WebVid-10M — watermarked stock footage, biased towards slow camera motion.

Both biases show in our own pilots. We are characterising this model, not video diffusion in general.

Vocabulary sampled from MSR-VTT and WebVid-10M captions; verbs from UCF101 and Kinetics-400 labels. We train on none of these.

Five cheap pilots, and four decisions that came out of them

Small and deliberately cheap. The point was to find out whether the project is tractable, not to produce results.

	THE QUESTION	WHAT WE RAN	WHAT WE SAW
E1	Can we generate video at all on our hardware?	AnimateDiff v1.5 through diffusers. 16 frames, 512×512, 25 DDIM steps, fp16.	**Runs.** 94 s per clip, 10.2 GB VRAM on a T4. This is the unit cost behind every budget in the plan.
E2	Does a fixed seed hold the scene still when one word changes?	Same noise tensor; prompt varied on one axis at a time (retriever→cat, running→walking).	**Yes, within a body plan.** Noun swaps shift appearance 2.6× more than verb swaps; layout SSIM 0.51 for horse against 0.26 for bird.
E3	Can we get activations out, and what does storage cost?	Forward hooks on the motion_modules and attn2; dumped shapes and sizes.	**Easy to hook: 95 coordinates, +0.6% runtime.** A full dump is 88 GB per clip, larger than the disk, so we store 45 MB of summaries.
E4	Do the temporal modules causally carry motion?	Scaled the motion-module residual towards zero, everything else identical.	**The clip goes near-static while each frame stays a plausible image.** The cheapest possible causal demonstration.
E5	Can we train a LoRA on this hardware, and at what clip length?	Stage 1, this week. One motion LoRA, 100 steps, rank 16, gradient checkpointing.	Pending. E1 set the inference budget before the sweep; E5 does the same for training.

Four decisions came out of this. AnimateDiff is primary (E1, E4) · every probing clip shares a fixed seed and noise tensor (E2) · the harness stores summaries, not raw tensors (E3) · content pairs stay inside a body-plan family (E2).

PART TWO

Five deliverables

Each one is only useful because of the one before it.



The trained model is numbered, not an afterthought: done the way Deliverable 5 describes, it tests Deliverable 2 rather than sitting beside it.

A full dump is larger than the disk, so we store summaries

01 **Hook both paths by name**

diffusers builds a UNetMotionModel from a UNet2DConditionModel plus a MotionAdapter. Temporal: the motion_modules in each down, mid and up block. Spatial: the transformer_blocks, attn1 and attn2.

02 **Index everything by one coordinate**

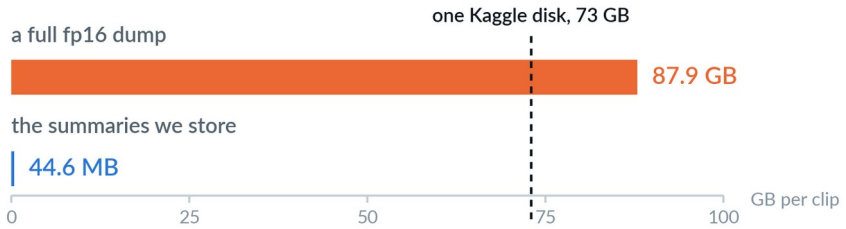
A triple of (module path, denoising timestep, clip id). Every later stage queries by that triple, so it is worth getting right once.

03 **Sweep in two tiers**

A coarse pass over 16 representative coordinates × 25 timesteps × 3 seeds, then a refined pass at full module resolution inside whichever block wins.

04 **Pin down determinism**

Each clip records its seed, sampler, step count, resolution, repository id and commit hash next to its activations.



2,017x smaller, per clip

31.6 TB -> 15.7 GB across all 360 clips

Measured on one clip: 95 coordinates, 25 timesteps, fp16.

What we keep per coordinate. Channel means and variances, the norms of the frame-mean-removed residual, and cross-attention maps head-averaged and downsampled to a 16 × 16 grid with the frame axis kept — for the subject, verb and scene tokens only. Full tensors are retained only where we flag a coordinate as interesting.

Still open: at 16 × 16 the attention maps are 78% of Kaggle's 20 GB output cap. An 8 × 8 grid halves the store. Decide before Stage 2 freezes the format.

Hooks and generation prototyped in E1 and E3. The production harness is Stage 2, 28 September to 4 October.

Change exactly one word — and hold the noise tensor too

The probe measures how two clips differ. If anything but the axis we care about also differs, the measurement is worthless.

the prompt template

a	[subject]	[verb]	[scene]	, high quality, detailed

a content pair

vary the subject, hold everything else

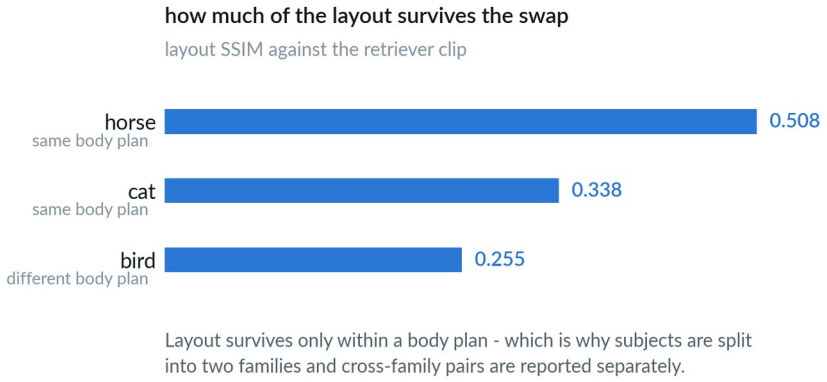
```
a golden retriever running on a beach
a horse running on a beach
```

a motion pair

vary the verb, hold everything else

```
a golden retriever running on a beach
a golden retriever walking on a beach
```

Same seed and the same noise tensor across a pair. Without that, the difference between two clips is mostly sampling noise and the probe measures nothing. E2 is what told us this is viable.



Pairs are labelled by construction

104 content pairs, 90 motion pairs, 56 cross-family. The anchor design fixes which pairs exist rather than leaving it to whichever we happened to use.

Three seeds, not one

42, 1337, 2024. A pair always lives inside one seed; the spread across seeds is the error bar on the map.

Vocabulary borrowed, not invented

Subjects and scenes from MSR-VTT and WebVid-10M captions; verbs from UCF101 and Kinetics-400 action labels.

Without the fixed noise tensor, the difference between two clips is mostly sampling noise. E2 is what told us this is viable.

Three predictions, each cheap to falsify

Sweeping blind produces a description. Going in with predictions produces a result.

H1 Layer axis

Motion selectivity peaks in the temporal modules at medium resolution, around up_blocks[1], rather than at the highest-resolution decoder layers or at the bottleneck.

Follows Xiao et al. Decided by argmax over blocks with a bootstrap CI; falsified if that interval reaches an extreme.

H2 Timestep axis

Motion is settled in roughly the first third of the denoising trajectory and appearance in the last. Content selectivity should run the opposite way.

Follows Baherwani et al. Decided by a changepoint fit with a bootstrap CI. A different boundary is a finding, not a bug.

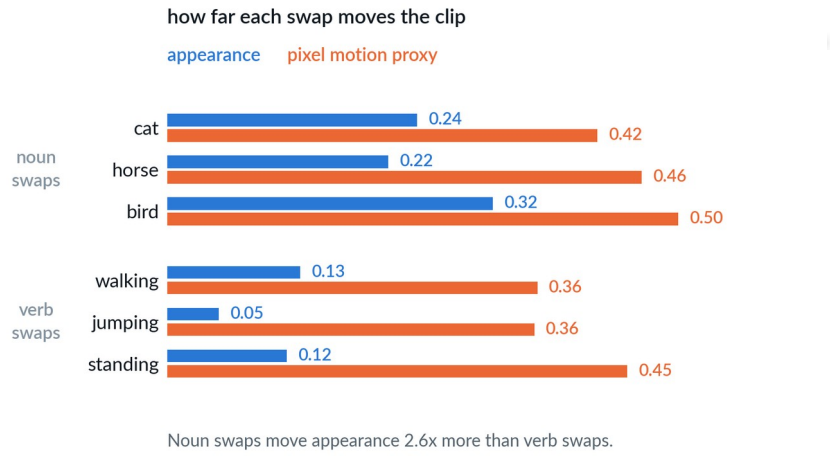
H3 The confound

Raw activations look content-selective everywhere. A clean motion signal only appears after subtracting the per-frame mean, which removes the static component.

The one that matters most. Skip it and we would wrongly conclude motion is nowhere.

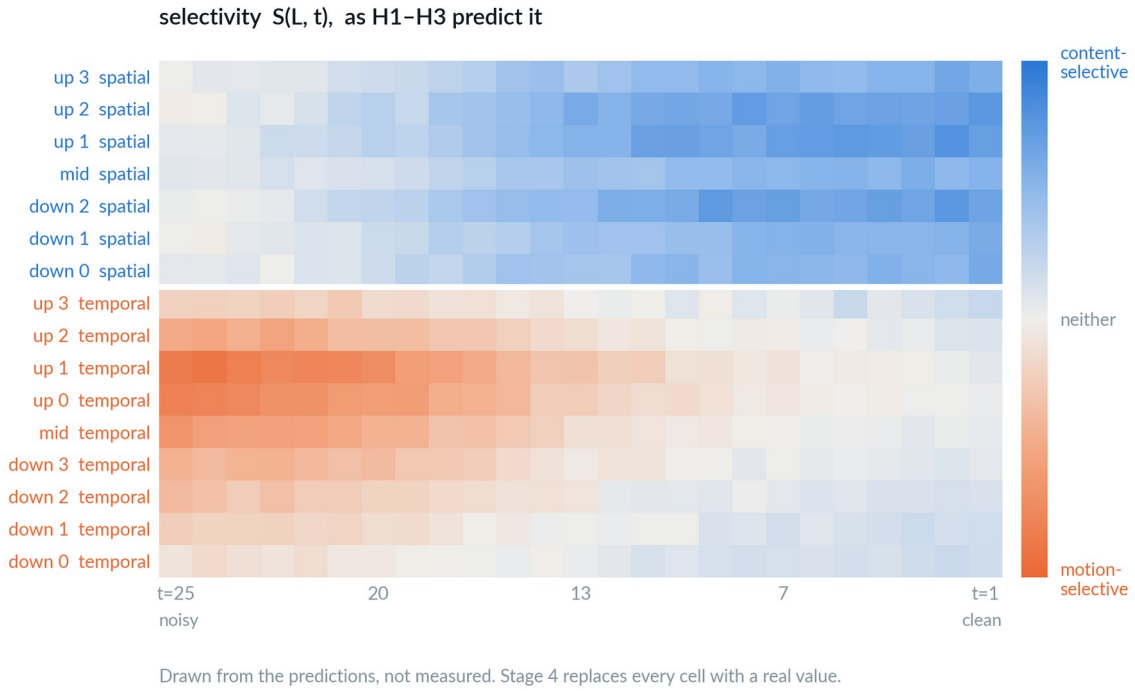
H3 is no longer only a prediction — E2 already measured it

Our raw pixel motion proxy read higher for noun swaps (0.46) than for verb swaps (0.39). Appearance swamped the motion measure before we touched a single activation — which is exactly why the map is computed on the frame-mean-removed residual, and why the static map is printed beside it as a control.



Noun swaps move appearance 2.6x more than verb swaps.

One number per layer and timestep, and its sign is the point



Drawn from the predictions, not measured. Stage 4 replaces every cell with a real value.

$$S(L, t) = \frac{d_c - d_m}{d_c + d_m}$$

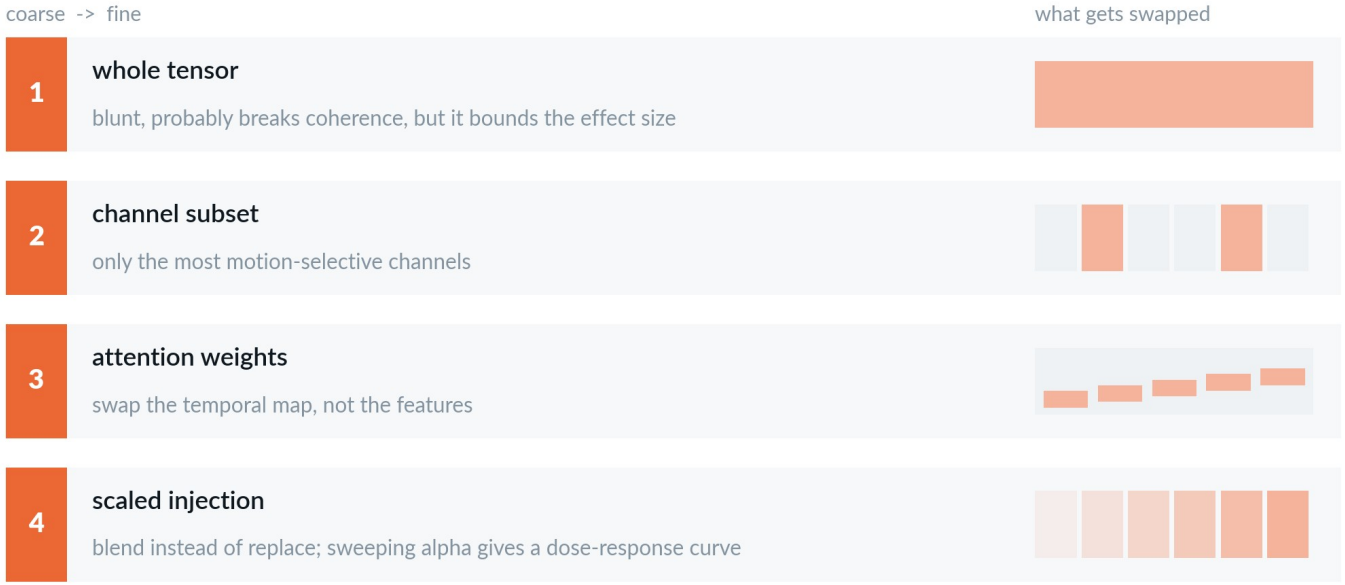
d_c is how far the activations move under a content swap, d_m how far under a motion swap, for a pair that shares a seed.

- **Positive** the coordinate tracks what is depicted
- **Negative** it tracks how things move

Distance is one minus cosine similarity, fixed before the sweep and not after seeing it. H3 makes this two maps: the dynamic one is the headline, the static one sits beside it as the control.

A second, independent view: a linear probe per coordinate says what information is present. Patching says what the model actually uses. Where they disagree is likely the most interesting thing we find.

Four granularities, and three controls we run first



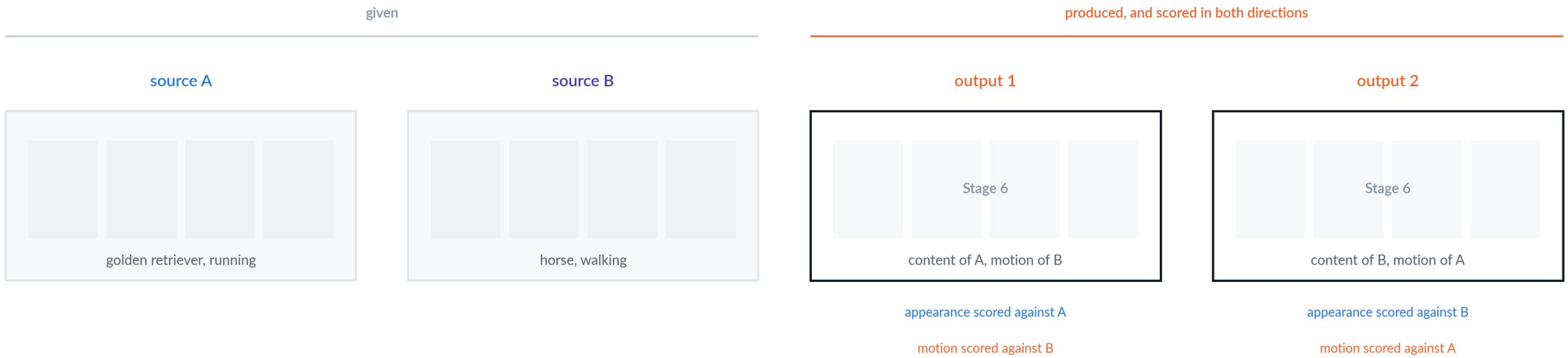
Controls we run first

- **Null patch**
  Patch a clip with its own activation. In fp16 nothing is bit-identical, so the bar is numerical: latent MAE below 1e-3, PSNR above 45 dB.
- **Random coordinate**
  Patch somewhere with low selectivity. If that works as well as our chosen coordinate, we have not localised anything.
- **Symmetry**
  A into B, then B into A. The two effects should be near inverses. Asymmetry means something is leaking.

Both runs must start from the same noise tensor, or the comparison means nothing. Corruption is resampling rather than noising, after Zhang and Nanda. Design is settled; implementation is Stage 5, 19 to 25 October.

The demonstration is a 2 × 2, and both diagonals have to work

Two edits per test case, each with an unambiguous target, across twelve test pairs in both directions.



A method that transfers motion but not content has found a useful lever, not a disentanglement.

The four placeholders above are replaced with real frames in Stage 6, 26 October to 1 November. Until then they are placeholders and we say so rather than showing something suggestive.

Alongside the grid we report the α sweep — the dose-response curve from granularity 4 — because a single hard swap hides whether the effect is graded or a cliff.

Train the motion LoRA twice — and the comparison is the point

two adapters, MotionDirector-style

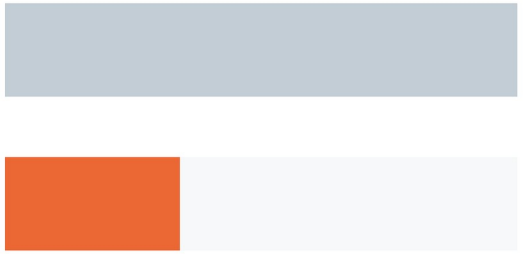
appearance LoRA

on the transformer_blocks attention projections, trained on one randomly sampled frame per step

motion LoRA

on the motion_modules, trained with the appearance LoRA attached and frozen

then train the motion LoRA twice



on every temporal projection

the full-coverage baseline

on the coordinates our map ranked highest

same rank, same data, a fraction of the trainable parameters

If the probe-guided one matches the full-coverage one, the map predicts in weight space and not only in activation space. If it does not, that is a real limit on the map -- and a better result than "we also trained a LoRA".

A third view, nearly free

‖ΔW‖ per module ranks layers by where training chose to put the motion.

Three rankings, one question

Patching says used, the linear probe says present, ΔW says learned. We report the rank correlation between all three.

Ordering is forced

Nothing starts before the Stage 4 map exists, because the probe-guided variant depends on it.

Cost and disclosure. Four training runs, about 6 GPU-hours, in Stage 6. The LoRA pair is fitted on roughly 20 real clips from UCF101 and Kinetics-400 — a bounded use of real video that touches nothing in Deliverables 1 to 4, and it is in the report's data section.

One headline number, three baselines, one sanity check

Content fidelity

CLIP similarity to the intended content prompt, plus CLIP or DINO image-to-image similarity against the reference frames. The image-to-image version is the honest one.

Motion fidelity

Trajectory-based, after Yatim et al.: point tracks from CoTracker, compared as displacement vectors over time. RAFT flow correlation as a cheaper second opinion.

Did we wreck the clip?

Two or three VBench dimensions — subject consistency, motion smoothness, temporal flickering. Cheap to run, and it answers the obvious objection before it is raised.

$$D = \tfrac{1}{2}[\, s_c(\text{out}, B) + s_m(\text{out}, A) \,] - \tfrac{1}{2}[\, s_c(\text{out}, A) + s_m(\text{out}, B) \,]$$

for an output meant to take B's content and A's motion

Both terms are min–max normalised across the evaluation set first: a CLIP cosine and a flow correlation are not commensurable. Raw values are printed alongside.

Baselines — without these, the number means nothing

Prompt-only

Same seed, new prompt, no patching. If we cannot beat this, we have built an expensive way to do nothing. Mandatory.

Training-free

MotionClone. Public code, comparable in spirit to what we are doing. Decided, not “or MOFT”.

Trained, our own

The probe-guided LoRA pair. Cheaper than running MotionDirector's pipeline and directly comparable.

Success: beat prompt-only in both directions with a bootstrap CI excluding zero, and stay within 10% of source on VBench subject consistency.

Twenty-two decisions, written down before the data arrives

A method whose choices are still open when the data lands is a method that can be fitted to whatever came out.

IMPLICIT BEFORE, FIXED NOW

1 The distance behind the map

One minus cosine similarity — on the frame-mean-removed residual for motion, on the frame mean for content.

2 Two maps, not one

S_dyn is the headline, S_stat is printed beside it. That is what H3 implies: demonstrate the confound instead of asserting it.

3 Pairs, seeds and vocabulary

Anchor design: 104 content pairs, 90 motion pairs, three seeds over the whole grid, enumerated in a config file.

THRESHOLDS, DECIDED IN ADVANCE

1 The null-patch tolerance

fp16 is never bit-identical, so “identical” needed a number: latent MAE below 1e-3, PSNR above 45 dB.

2 D was not commensurable

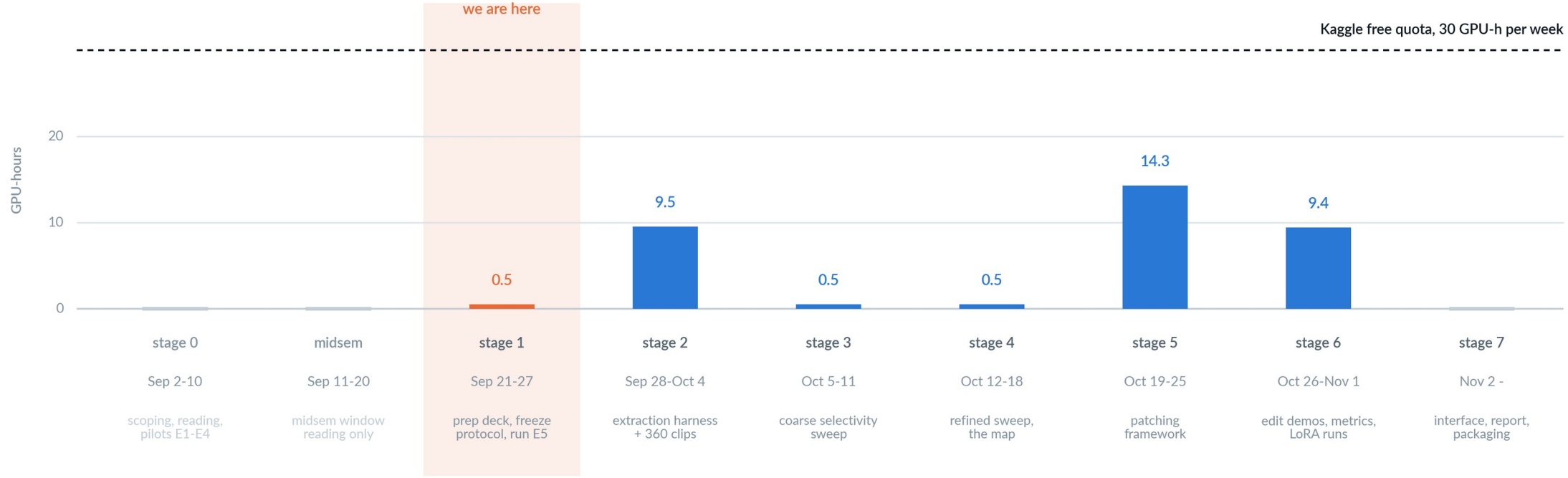
It added a CLIP cosine to a flow correlation. Both terms are now min–max normalised, with raw values printed alongside.

3 What would count as success

Beat prompt-only in both directions with a bootstrap CI excluding zero, within 10% of source on VBench subject consistency.

All twenty-two are listed in §5.7 of the roadmap, dated and append-only, so a choice made in September is not quietly remade in October to suit the data.

Eight stages, and Stage 2 is the one most likely to slip



Where the slack is

Video inference and hook-based extraction are slower and more fragile than the image-diffusion equivalent. If Stage 2 overruns, the time comes out of Stage 5, not Stage 3: a rushed patching implementation is recoverable, a rushed sweep is not.

Why the totals carry ±10%

E1 measured 94 s per clip with a within-session SD of 0.1 s, but throughput varies about 10% between Kaggle sessions — E3's no-hook baseline on a later session read 85.1 s. Hence 34.7 ± 3.5.

What the tiering bought

A naive sweep of 9 layers × 7 timesteps × 12 pairs × 2 directions is 1,512 generations, 39 GPU-hours on its own. The two-tier plan exists because of that arithmetic, not for elegance.

Four things that could sink this, and what we do about each

Compute limits the layer × timestep sweep	→	Start coarse — 16 representative coordinates, then refine only around what looks promising. E1 measured 94 s per clip and E3 adds only 0.6% for hooks; the tiered plan totals 34.7 GPU-hours against a 30 GPU-h weekly quota, where the naive sweep alone would have needed 39.
AnimateDiff's split is too easy to be a finding	→	Fair criticism, and we say it up front. The answer is ModelScope T2V, where spatial and temporal were trained together. The probe-guided LoRA is the cheaper second answer: it asks whether the map predicts anything, not whether it echoes the architecture.
Patched outputs come out incoherent rather than wrong	→	Run the null patch and the random-coordinate control before believing anything. Prefer scaled injection over hard replacement, and report the whole α curve rather than one point on it.
Everything looks content-selective and we find nothing	→	This is H3, and it is the failure mode we most expect. Subtract the per-frame mean before scoring. E2 already showed the confound at pixel level; reading Xiao et al. saved us a week here.

Open questions for the TA are listed in the roadmap document, including what counts as an acceptable interface for this project.

References · 1 of 2

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References · 2 of 2

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Thank you

Questions and discussion

Team Claudn't

Roll numbers — [fill in, one per team member]

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